

Engineering Reboot 2026

Week 02 - Day 08 Field Manual v3.1

Loops and Repetition

PHASE 0 – MISSION BRIEF

Objective

Learn how computers perform repetitive tasks using loops.

Core Concepts

- for loops
- range()
- counting
- iteration
- accumulation
- loop boundaries

Engineering Applications

- Production counters
- Test automation
- Inventory processing
- Data analysis
- Embedded systems

LEARNING OBJECTIVES

By the end of Day 08 you should understand:

- for loops
- range()
- loop iteration
- counting sequences
- accumulation
- repetitive execution

DELIVERABLE INVENTORY

1. counting_program.py
2. even_number_generator.py
3. sum_calculator.py
4. Notes.md
5. Journal.md
6. Evidence Package

PHASE 1 – COUNTING PROGRAM

Implement

- Use a for loop
- Use range()
- Display numbers 1 through 10

Validation Matrix

1
2
3
4
5
6
7
8
9
10

Evidence Checkpoint

- counting_program-Code.png
- counting_program-Output.png

Checkpoint

- ☐ Validation complete
- ☐ Evidence captured

PHASE 2 – EVEN NUMBER GENERATOR

Implement

- Use a for loop
- Use range()
- Display only even numbers

Validation Matrix

2
4
6
8
10
12
14
16
18
20

Evidence Checkpoint

- even_number_generator-Code.png
- even_number_generator-Output.png

Checkpoint

- ☐ Validation complete
- ☐ Evidence captured

PHASE 3 – SUM CALCULATOR

Implement

- Use a for loop
- Use an accumulator variable
- Display final result

Validation Matrix

Expected Result: 5050

Evidence Checkpoint

- sum_calculator-Code.png
- sum_calculator-Output.png

Checkpoint

[] Validation complete

[] Evidence captured

PHASE 4 – NOTES

Create Week-02/Notes/Day-08/Notes.md

Day 08 Notes

Concepts Learned

Key Commands

Key Observations

Lessons Learned

Questions For Future Study

[] Notes completed

PHASE 5 – JOURNAL

Create Week-02/Journal/Day-08/Journal.md

Day 08 Journal

Summary

Challenges Encountered

Solutions Applied

Confidence Assessment

Reflection

Tomorrow Preparation

[] Journal completed

PHASE 6 – GIT OPERATIONS

Execute:

git status

git add .

git commit -m "Complete Day 08 loop exercises"

git push

git fetch

git status

git branch -vv

Preferred End State:

Your branch is up to date with 'origin/main'

nothing to commit, working tree clean

Evidence Checkpoint

- Git-Operations.png

Must show:

- git add .
- git commit
- git push
- git fetch
- git status
- git branch -vv

END-OF-DAY SUCCESS CRITERIA

- ☐ counting_program.py completed
- ☐ even_number_generator.py completed
- ☐ sum_calculator.py completed
- ☐ All validation completed
- ☐ Exercise evidence captured
- ☐ Notes completed
- ☐ Journal completed
- ☐ Git workflow completed
- ☐ Git-Operations.png captured
- ☐ Remote synchronization verified
- ☐ Working tree clean

MISSION COMPLETE ✓

DAY 09 PREVIEW

Tomorrow's Topic
Functions

Expected Deliverables

- greeting_function.py
- calculator_function.py
- temperature_converter.py

Core Concepts

- def
- parameters
- return values
- function calls
- code reuse